



Green growth and sustainable development: dynamic linkage between technological innovation, ISO 14001, and environmental challenges

Nabila Abid¹ · Federica Ceci¹ · Muhammad Ikram² 

Received: 2 July 2021 / Accepted: 9 November 2021

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2021

Abstract

Over the last two decades, the phenomenon of green growth has gained much attention from academics and policymakers striving to find a sustainable solution to environmental problems worldwide. Technological innovation serves as a tool to abate the acute environmental crisis and continuously promote sustainable development by converting traditional economies into green economies. Pakistan is among the developing countries relying on conventional technology and energy resources to meet population and economy demands, which has led to a surge in greenhouse gases and other hazardous air pollutants. The literature exploring green growth in the Pakistani context is scant; the present study will therefore fill this gap and explore the dynamic linkage between technological innovation and ISO 14001 with green growth in Pakistan in the presence of environmental challenges such as energy consumption and population growth. A novel grey relational analysis model approach is employed to examine the interrelationship between the study parameters. Results indicate that technological innovation is significantly correlated with green growth, and ISO 14001 also shows a substantial relationship with green growth. However, among the environmental challenges, energy consumption poses a barrier to green growth development as the country's energy mix is dominated by fossil fuels as compared to renewables. The research findings produce a much-needed policy suggestion to address the environmental challenges by promoting green growth development in Pakistan.

Keywords Green growth · Sustainable development · Technological innovation · ISO 14001 · Environmental challenges

Introduction

Since the industrial revolution, the choice of energy sources to spur economic growth has been criticized. The resulting ecological crisis has made it one of humanity's most pressing challenges in the twenty-first century. The excessive use of fossil fuels to meet economic and social needs has resulted in surging emissions of CO₂ and other greenhouse gases (GHGs). The dramatic change in environmental conditions is a daunting concern for all academics and

policymakers worldwide, who are looking for an optimal solution to the mentioned crisis.

In 2005, the United Nations proposed green growth as the ultimate solution to achieve sustainable development. The Organisation for Economic Co-operation Development (OECD) defines green growth as a new means of fostering development and economic growth in a country while causing no harmful impact on the environment or people living in it (Merino-Saum et al. 2020; Wang et al. 2020). Green growth has drawn the attention of the masses around the world, and a plethora of research has been conducted to understand the concept of green growth in different countries across the globe (Bagheri et al. 2018; Stoknes and Rockström, 2018; Wang and Shao 2019). Green growth provides a solution to the environmental crisis and helps achieve sustainable economic growth while flourishing the interests of the population (Barbier 2016). Green growth promotes cleaner energy sources and reduces environmental pollution by lowering the emission of CO₂ and other GHGs (Sandberg et al. 2019). Several authors have argued about multiple

Responsible Editor: Eyup Dogan

✉ Muhammad Ikram
i.muhammad@aui.ma

¹ Department of Economia Aziendale, University G. d'Annunzio Chieti, Pescara, Italy

² School of Business Administration, Al Akhawayn University in Ifrane, Avenue Hassan II, P.O. Box 104, 53000 Ifrane, Morocco

determinants of green growth, but have agreed on all factors that help develop and sustain economic, environmental, and social interests (Hao et al. 2021; Mensah et al. 2018).

The industrial revolution brought tremendous changes in the way of doing things. To date, most developing countries have relied on traditional energy sources to meet national demands, resulting in serious environmental issues (Tugcu et al. 2012). Energy production from fossil fuels is a dominant factor in escalating CO₂ emissions globally (Köne and Büke 2010). CO₂ and GHG emissions are the byproducts of energy combustion from conventional energy sources like coal, gas, and oil (Wang et al. 2020). These emissions threaten human lives (Alam et al. 2011; Osobajo et al. 2020). Therefore, reducing carbon emission to abate global environmental catastrophes has become a significant concern worldwide (Acosta et al. 2020).

Technological innovation is a tool to address environmental challenges, reduce the potential climate change threats posed to human lives, and sustain economic interests (Li and Long 2020). Technological innovations have played a crucial role in transitioning society from a traditional economy to a green one. Technological innovation supports green energy production so as to channel the development of green growth and help transform technological practices in operations and production processes within organizations (Aghion et al. 2013; Umar et al. 2020). According to Song et al. (2019), transformative technology innovation helps meet national energy demands by promoting cleaner technologies that reduce CO₂ and other harmful emissions. Wang et al. (2021) suggested that technological innovation acts as a catalyst to boost green growth. In addition to technological innovation, some organizational innovations, such as Environmental Management System (EMS), can help firms reduce the impact of their production. EMS is one of the widely adapted environmental sustainability tools that help public or private organizations engage in systematic processes to tackle and solve environmental issues and lower harmful impacts (Coccia and Watts 2020; Rusko et al. 2014). The research conducted by Kammerer (2009) reveals that green practices (drawn from EMS) and innovations within an organization share a positive relationship. Prior studies have shown that EMS promotes environmental sustainability and performance by involving managers in the process (Aravind 2012; Mungai et al. 2020).

Pakistan is a developing country, and despite having potential for renewables, its government has relied heavily on fossil fuels and traditional technological methods to meet its economic and social needs (Hussain et al. 2019). The International Energy Agency (IEA) rated Pakistan as the fifth most vulnerable country to environmental issues (International Energy Agency (IEA), 2018). Shahzad et al. (2017) found that the dependence on scarce energy resources will have a prolonged impact on the country's environment

and human lives. Substantial CO₂ emissions have posed a tremendous environmental challenge for Pakistan, imposing considerable threats to the environment. While PM 2.5 (particulate matter) is the most dangerous particulate matter in the air, drastically impacting human lives, it is a byproduct of fossil fuels released from power plants and vehicles. PM 2.5 is a heavy air pollutant responsible for severe lung and heart conditions (Wu 2017). According to the World Health Organization (WHO), more than 40 million cases and 28,000 deaths due to severe lung and respiratory diseases are reported annually in Pakistan (Environmental health 2020). Abid et al. (2020) argued that in recent years, conventional energy consumption and population growth have caused several environmental problems, and under Belt and Road energy projects, Pakistan would produce 67% of its energy from fossil fuels like coal, gas, and oil. If technology, process, and manufacturing all remain the same, the region will face significant environmental disasters. The industrial sector in Pakistan relies on energy from fossil fuels and has become the second significant contributor to GHG emissions in Pakistan (Shrestha and Shrestha 2013). The roots of the environmental problem cannot be traced to energy choices in Pakistan, as Naqvi and Rehm (2014) found that with every 1% increase in population, energy consumption and emissions would increase by 0.46% in the country.

In the light of the above, building upon secondary data collected in Pakistan and an extensive literature review, this study aims to answer a key research question: Do technological innovation and EMS adaption help the development of green growth in Pakistan?

Research concerning technological innovation and green growth is scarce (Popp 2012). In the literature, the phenomenon of green growth has not been widely explored in the Pakistani context. Therefore, this is one of the most significant contributions in the field as Pakistan is a rapidly developing country that relies heavily on fossil fuels to meet its demands to spur economic growth (Abid et al. 2020). This study fills the gap in the literature and provides a direction for future studies revolving around green growth development in the country and offers policy implications related to green growth for designing relevant policies that shape the future of sustainable development for the country. This study extends its contribution to the literature by examining the integrated effect of technological innovation with the development of green growth in Pakistan.

Furthermore, this study proposes a more comprehensive framework by combining technological innovation and EMS (organizational innovation) with two environmental challenges (i.e., energy consumption and population growth). To do so, the data of technological innovation (R&D, patent applications), EMS (organizational innovation), and green growth (CO₂ emissions, renewable energy, GHGs, and PM 2.5) from 2000 to 2019 are used. This study employs the

novel GRA model approach, offering consistent, unbiased solutions as this technique offers optimal results even with a small data sample and, as a result, identifies the most effective element to promote green growth in Pakistan. The main advantage of GRA models is to ensure accuracy and superiority over statistical models (Ikram et al. 2020). Lastly, this study can be generalizable to economies having similar economic and social contexts.

The rest of the paper is structured as follows. The “Literature review” section presents the extensive literature on technological innovation, EMS, and green growth. The “Research methodology” section describes data information and the steps involved in the construction of GRA models. The “Results and discussion” section provides the results and discussion. The conclusion, policy implications, and future directions are presented in the last section of the study.

Literature review

Technological innovation and green growth

For over a century, increased resource consumption (fossil fuels, water, forests) has spurred growth. Rapid economic developments and increased human activities have been fueling environmental problems for decades leading to ecological catastrophes (droughts, hurricanes, and global temperature rise) that are not limited to national boundaries (Kunapatarawong and Martínez-Ros 2016; Tseng et al. 2013). Among many, the most pressing environmental challenges are increasing CO₂ emissions, which are a byproduct of the overuse of nonrenewable energy production sources. Nonrenewable energy sources are scarce, and economies have been utilizing these resources for years, which has accelerated the growth process but adversely affected the environment (Du et al. 2020). Environmental challenges have a detrimental impact on human health, and continued degradation will jeopardize long-term economic goals (Xu et al. 2021). Development at the expense of the environment has alarmed global leaders who are now striving to find a sustainable solution to tackle the environmental crisis worldwide (Cuerva et al. 2014; Li et al. 2020).

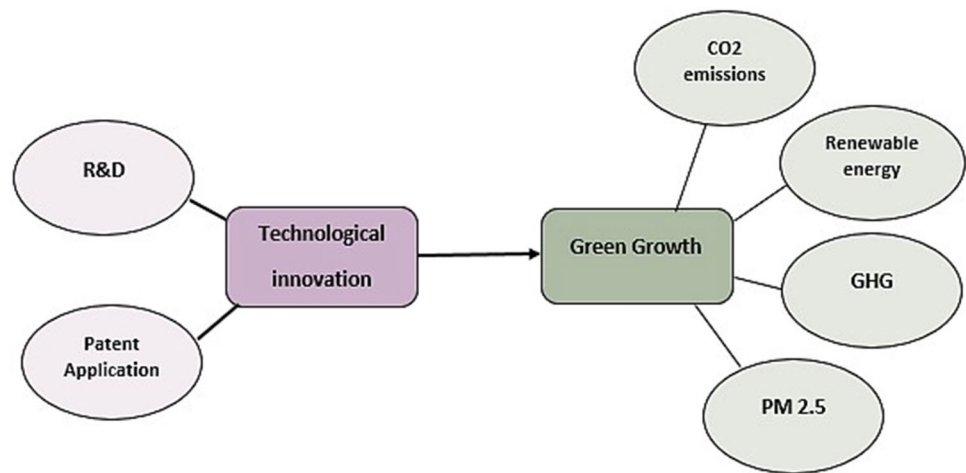
In 2005, the United Nations proposed green growth as a means to achieve sustainable economic growth without harming the environment (Kang 2015). Green growth promotes the nationwide pursuit of growth and development while ensuring the efficient use of natural resources to protect the environment (OECD 2011). Therefore, the transformative characteristic of green growth has drawn global attention. “Green” represents harmony between humans and the environment, and “growth” encompasses the prosperity of a country and its people. Green growth provides an ultimate solution to the environmental challenges and

accelerates the deployment of clean energy sources to reduce environmental degradation (Aguilera-Caracuel and Ortiz-de-Mandojana 2013; Li et al. 2019a).

Scholars (Hao et al. 2021; Su et al. 2020) have recently emphasized the role of technology in promoting green growth development. However, the effect of technological innovation on green growth is much debated in the field. Technology is one of the prime factors that organizations have been using to excel in performance and constantly impact humans and the environment (Li et al. 2019b). Environmental awareness has made policymakers realize that technology intensifies environmental issues, and progress leading to environmental catastrophes is not advisable. Therefore, a shift has been observed to upgrade traditional technology to green or clean technology through resource-efficient innovation and to substantially reduce environmental pollution (Schiederig et al. 2012). Green innovation is divided into four categories: managerial, product, process, and technological innovation (Tseng et al. 2013). Technological innovation preserves environmental interests through the deployment of green technology that accelerates the process of green growth, which ultimately reduces the potential threats of climate change to human lives (Li and Long 2020). Johnstone et al. (2010) suggested that technological innovation is imperative for green growth development. Danish and Ulucak (2020) examined how technological change affects green growth in BRICS countries. Using CO₂ as a proxy for green growth, the authors argued that technology is critical in green growth transition. Ahmad et al. (2020) considered technological innovation to be an essential determinant of green growth. They stated that technological innovations transform traditional technology into a cleaner one to protect the environment and improve operational efficiency. This study is endorsed by previous research by Aghion et al. (2016) who argued that technological innovation should be blended into policy plans to promote green growth. Figure 1 presents the relationship between technological innovation and green growth.

Studies have estimated technological innovation using R&D funding and patent applications (Zhai and An 2021). Aghion et al. (2005) used patent application as indicators of technological innovation. Berghäll (2016) measured R&D as a representative of innovation. The results of the study by Audretsch and Belitski (2020) and Shaikh and Colarelli O'Connor (2020) showed that R&D is vital to innovation in an organization and improves productivity. The authors further stated that the impact of R&D may differ across countries and depends on the variables under study. According to Wang and Zhu (2020), technological innovation is essential to reduce carbon dioxide emissions. R&D funding and investments promote technologies, reducing CO₂ and other GHG emissions (Jung et al. 2012; Sohn et al. 2015). According to Fernández et al. (2018), R&D funding is effective

Fig. 1 Technological innovation and green growth



in minimizing CO₂ emissions. Song et al. (2019) stated that not only does technological innovation meet national energy demands by promoting cleaner technologies, but it also reduces CO₂, GHG emissions, and other harmful pollutants that impose threats on human lives (Bekun et al. 2019).

In contrast to the above literature, several studies (Fisher-Vanden and Wing 2008; Wang et al. 2014; Zhao et al. 2017) have claimed that technological innovation enhances production capacity that requires more energy consumption, resulting in harmful emissions. Furthermore, Chen and Lee (2020) and Samargandi (2017) disclose that technological innovation does not impact global CO₂ emissions and does not significantly protect the environment. Therefore, this inconsistency within the literature motivates us to extend our analysis. For decades, technological innovation has been considered as a core aspect of national growth. However, the literature regarding technological innovation and green growth is limited (Popp 2012). Therefore, based on the aforementioned contradictory literature and with purpose of exploring the impact of technological innovation on green growth in the context of a developing country, we propose the following hypotheses:

Hp1: Patent application is positively related to green growth.

Hp2: R&D is positively related to green growth.

Green organizational innovation, ISO 4001, and green growth

Daunting environmental challenges call for organizational support to preserve and protect the environment. Environmental Management Systems (EMSs) are the environmental sustainability tools that help organizations address environmental challenges and ensure environmental sustainability. A number of EMSs exist, but ISO 14001 is the most adapted management standard worldwide. ISO 14001 was

established in 1996 by the International Organization for Standardization (ISO), and it has been revised over the years. ISO 14001:2015 is the latest version with a set of updated requirements to improve environmental performance (da Fonseca 2015). After ISO 14001, the second most famous EMS standard is the Eco-Management and Audit Scheme (EMAS) designed and established by the European Regulation EC 1221. EMSs make organizations follow specific rules to handle environmental management aspects both internally and externally (Chiarini 2017; Testa et al. 2014).

According to Könnölä and Unruh (2007), EMS is a substantial part of managerial innovation, ensuring efficiency in process control, production improvement, and product quality. Hence, EMS can be seen as a management standard applied to the whole industrial network. Curkovic and Sroufe (2011) and Ikram et al. (2020) argued that EMS is not just a part of environmental improvement; an organization can also use EMS to create a positive public image and involve managers from different hierarchical levels to engage in innovation. Similarly, the said EMS is a key instrument introduced to promote product innovation. The research conducted by Kammerer (2009) reveals that green practices (drawn from EMS) and innovations in an organization share a positive relationship. EMS ensures environmental efficiency and enhances organizational innovation capabilities (Darnall 2006).

This study mentions and employs ISO 14001 because EMAS has not yet been implemented or established in Pakistan. ISO 14001 serves as a part of the Environmental Management System and involves the planning, coordination, processing, resource deployment, implementation, and evaluation of an environmental policy (Darnall et al. 2008). In the last two decades, a global trend to adopt EMS has been observed. EMS has proven to be efficient in addressing environmental issues and increasing organizations' accountability in considering environmental concerns (Mazzi et al. 2017). ISO 14001 certifications help organizations achieve

environmental goals and provide a way to design and manage organizational operations with superior environmental performance (Carrillo-Labela et al. 2020).

Aravind (2012) argued that in order to implement proper practices to improve environmental performance in an organization, the management enhances process innovation and environmental audit. To obtain ISO 14001 certifications, organizations need to comply with ISO standards and requirements. First, organizations must develop an environmental report mentioning all the potential activities, impacts, goals, and plans to minimize environmental impact. Second, organizations must disseminate information about ISO 14001 with employees at different levels, train them, and systematically handle environmental issues. Henriques and Sadorsky (2007) and Naveh et al. (2006) define ISO 14001 as an administrative innovation within an organization that promotes process-oriented practices. These innovations facilitate organizational change in management practices and operations (Walker et al. 2011). According to Wagner (2010), understanding the ISO 14001 as a measure of organizational innovation is not simple. Aragón-Correa and Rubio-López (2007) and Ziegler and Seijas Nogareda (2009) argued that ISO 14001 increases environmental performance by reducing harmful impacts and improves organizational performance by channeling innovation capabilities. EMS drives innovation capabilities in management practices and environmental processes (Bansal and Hunter 2003). The advantages of ISO 14001 compel multiple economies to support EMS deployment by providing financial and technical support to organizations (Könnölä and Unruh 2007).

EMS helps organizations consistently and systematically tackle significant environmental issues, including air pollution. Countries must facilitate EMS adoption if they want to reduce air pollution (Raines 2002). The updated ISO 14001:2015 incorporated various aspects of energy

consumption and emphasized the use of process, operational techniques, and resources that do not discharge harmful air pollutants, emissions, or other residual waste to avoid environmental impacts. ISO:14,001:2015 thereby supports the goal of green growth by stressing the need for clean energy sources/materials and techniques (technologies) to minimize drastic emissions and avoid harmful impacts on the environment (Laskurain et al. 2017).

The literature shows the importance of EMS in environmental protection and sustainable development, but despite adopting ISO-14001, contradictory results worldwide make it controversial. There is no globally unified EMS result (Boiral and Henri 2012; D'Souza et al. 2019), which has forced this study to incorporate EMS into its framework and verify its role in green growth development in Pakistan with other variables. There is a direct relationship between EMS and green growth, and it is presented in Fig. 2.

Therefore, we propose the following hypothesis:

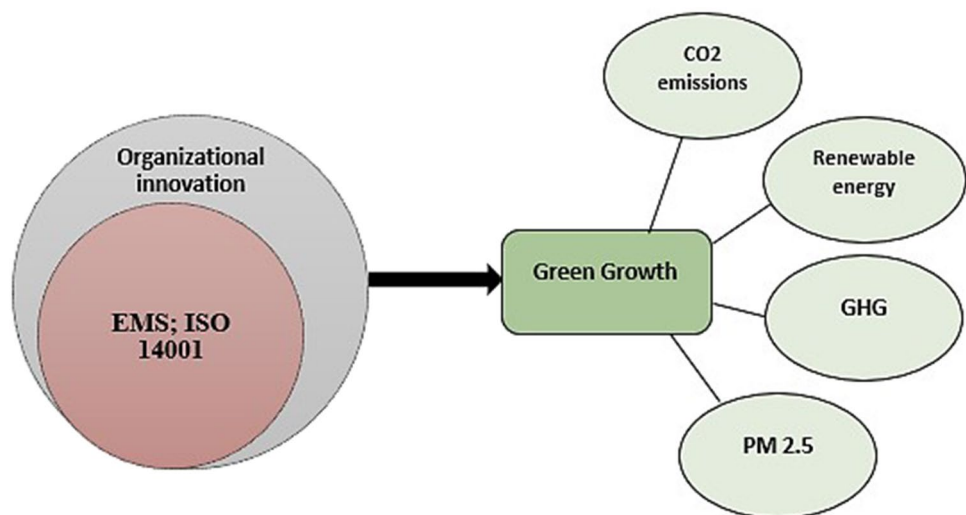
H_{p3}: Green forms of organizational innovation (i.e., EMS) are positively related to green growth.

Environmental challenges and green growth

Energy consumption has contributed to the growth of economies around the world, but it has also damaged the environment over the years. Energy consumption has played a substantial role in the global environmental crisis (Sato 2007). Alkaher and Carmi (2019) stated that the environmental catastrophe is a byproduct of human activities worldwide. They further argued that population growth is among the most significant causes of the environmental crisis, which should be addressed next.

This study incorporates the two most significant environmental challenges into its framework; the first environmental

Fig. 2 EMS and green growth



challenge is represented by energy consumption, and the second is population growth, presented in Fig. 3. Since the industrial revolution, fossil fuels have been the dominant driver of world economies used to enhance production and meet the increasing population demands. Increased energy consumption leads to soaring emissions of CO₂ and other GHGs (Hamit-Haggar 2012). GHGs are considered to be the main factor responsible for global climate change and a significant threat to the environment and humanity (Davis et al. 2018; Miao 2017).

The primary source of pollution is energy consumption from traditional energy sources like coal, gas, and oil in various industries, including energy and transportation (Odeh & Cockerill 2008). Human activities have increased the level of air pollution (Feng et al. 2019). Air pollution is a threat to human lives and other species sharing the same environment (Ruggieri and Plaia 2012). Besides CO₂ and GHG emissions, PM 2.5 is the most dangerous particulate matter in the air. The most harmful impact on human lives is a byproduct of fossil fuels released from power plants in factories and vehicles. PM 2.5 is a heavy air pollutant responsible for severe lung and heart conditions (Wu 2017). Previous studies have shown that GHGs and PM 2.5 are the result of the overuse of fossil fuels and mass industrial production (Kantor et al. 2010).

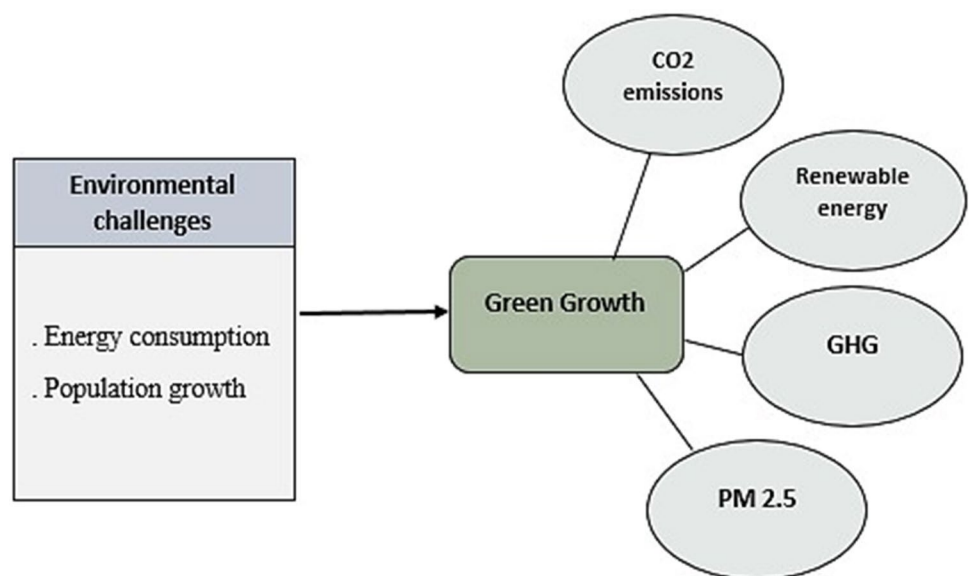
Zhang et al. (2021) suggested replacing fossil fuels with renewable energy sources to minimize harmful gas emissions. One of the best strategies to reduce CO₂ emissions is to prefer renewable energy consumption. Their use will protect the environment and human species from harmful emissions and serve as a tool to achieve sustainable development (Sandberg et al. 2019). In their studies, Danish and Ulucak (2020) and Odeh and Cockerill (2008) emphasized the use of renewable energy consumption and argued that

the establishment should help the public and private sectors reduce their reliance on conventional energy. Most of Pakistan's energy consumption comes from fossil fuels. The combustion of fossil fuels has caused a surge in CO₂ emissions in Pakistan, thus resulting in a prolonged environmental crisis (Latif et al. 2014).

Population growth represents the second dimension of environmental challenges that we included in our research. Since the industrial revolution, population growth and excessive energy consumption produced from fossil fuels have disturbed the environment and posed a threat to younger generations (Smith et al. 2013). Air quality represents environmental quality; thus, environmental crisis and air quality are interlinked (Roberts and Wooster 2021; Shi and Wu 2020). Javed et al. (2019) disclosed that in developing economies, the leading cause of CO₂ emission is population and traditional energy sources of production. The research findings of Sulaiman and Abdul-Rahim (2018) showed that population growth is one of the most significant drivers of CO₂ emissions.

Rehman et al. (2020) reported that energy consumption is the primary cause of emissions, but population growth is also causing CO₂ emissions as humans are natural emitters of carbon dioxide. Cui et al. (2018) analyzed various social and economic indicators and found that population growth increases CO₂ emissions. Naqvi and Rehm (2014) and Abid et al. (2020) examined Pakistan's socioeconomic and environmental variables and revealed a positive relationship between population growth and CO₂ emissions. With every 1% increase in population, there will be a 0.46% increase in CO₂ emission, a significant reason behind its deforestation, and fewer initiatives to protect and sustain Pakistan's natural environment. Uncontrolled population growth demands more energy and sources and exerts pressure on

Fig. 3 Environmental challenges and green growth



the environment, and in the absence of proper precautionary measures, the ecological system will suffer.

Therefore, we propose the following hypotheses:

Hp4: Energy consumption is negatively related to green growth.

Hp5: Population growth is negatively related to green growth.

This study develops a multi-framework by incorporating technological innovation, green organizational innovation, and environmental challenges with green growth. It expands the range of desirable outcomes responding to Pakistan's green growth development and environmental goals. We aim to test the impact of independent variables on green growth by employing the GRA approach. The analytical model summarizing all the hypotheses and variables used in the study is reported in Fig. 4.

Research methodology

This study aimed to explore the potential impact of technological innovation, EMS, and environmental challenges (i.e., energy consumption and population growth) on green growth in Pakistan. In order to gain a better understanding of technological innovation, the study used research and development (R&D) and patent applications (PA) as a proxy of technological innovation (Aghion et al. 2005; Berghäll 2016). EMSs such as ISO 14001 certifications were

considered since Pakistan has not implemented EMAS yet. ISO 14001 represents the number of organizations in Pakistan that have implemented EMS. Energy is measured in terms of energy consumption and population as the percentage of population in Pakistan. Green growth is measured in terms of air pollution, with PM 2.5 (particulate matter) considered as the most dangerous air pollutant and a significant threat to human lives (Diao et al. 2019), GHGs, and renewable energy consumption (REC), which is the crucial strength factor to produce clean energy and promote sustainable development (Abid et al. 2021). Data from 2000 to 2019 were used, and a complete description of variables and data collection sources is reported in Table 1.

Grey relational analysis (GRA)

The grey relational model, known as the GRA model, has received enormous recognition in the past two decades. GRA models are applied in different fields of study, including management, engineering, economics, environmental science, and medicine (Xu and Li 2018). GRA methods of data analysis have emerged as an attractive option as they provide an advantage over other techniques in several ways. GRA allows for efficient handling of small datasets, dealing with grey values, and providing optimal results (Ikram et al. 2021a).

In the present study, GRA models consist of four stages (Fig. 5). For the analysis, data were extracted from multiple databases from 2000 to 2019 and normalized in the first stage. In the second stage, GRA models were run

Fig. 4 Conceptual model

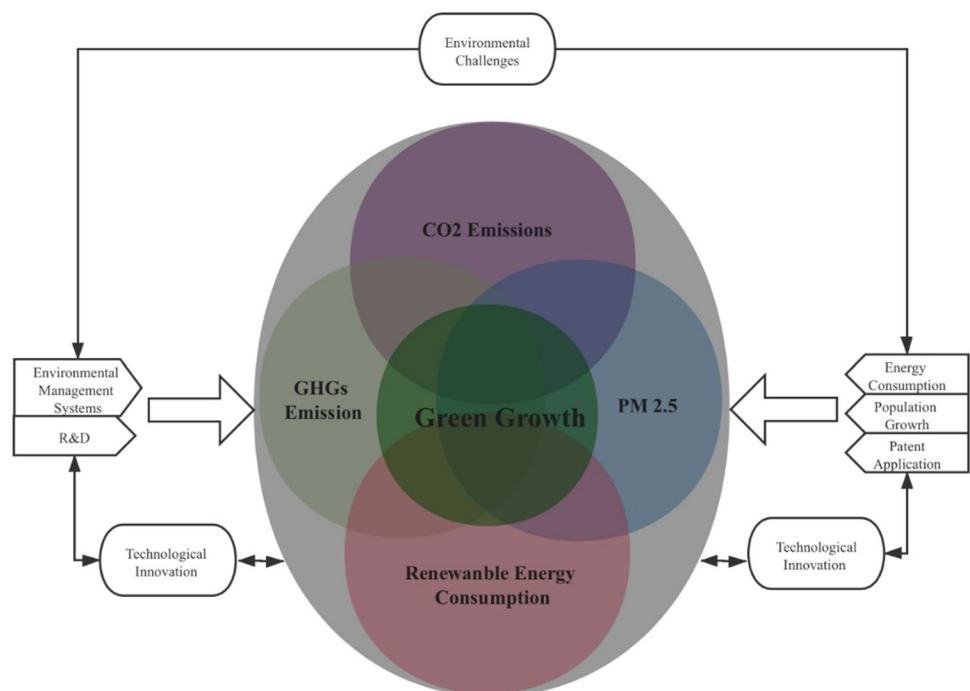


Table 1 Description of study variables

Variables (code)	Description of variables	Measurement	Data sources
R&D	R&D expenditure	Research and development expenditure, percent of GDP	Database of business and economic data
PA	Patent applications	Number of patent applications	WGI
EC	Energy consumption	Energy consumption in the industrial sector	Our World in Data
EMS	EMS	ISO 14001 certifications	ISO
PG	Population growth	Population growth	WGI and database of business and economic data
REC	Renewable energy consumption	Percentage of renewable consumption (TWh)	Our World in Data
GHG	Greenhouse gas emissions	GHG in the agricultural sector (%)	WGI
CO2	Carbon dioxide emissions	Annual CO ₂ emissions (mt)	Our World in Data
PM 2.5	Air pollution/particulate matter 2.5	Mortality from exposure to ambient PM 2.5 (per 1,000,000 inhabitants)	OECD

on the data to examine the relationship that technological innovation (PA and R&D), energy, environmental management systems (EMSs) (ISO 14001), and population growth (PG%) have with green growth (GHG emissions, PM 2.5, CO₂ emissions, and renewable energy) in Pakistan. The third stage involved the use of the GRA ranking approach, employing the weights generated in stage two and, after defining the decision criteria, assigning them to the alternatives. In the last stage, the conservative model was run on the analysis results to find the best solution (the variable or set of variables), having a substantial impact on green growth development in Pakistan.

In 1982, Deng developed the grey system theory (Ju-Long 1982). GRA models, also known as grey incidence analysis (GIA), are considered to be the key elements of Deng's grey system theory (Badri Ahmadi et al. 2017). These models are superior to other approaches in tackling data issues, whether they contain missing values or unreliable data sources (Ikram et al. 2019). They also assist in handling small datasets with maximum efficiency (Rehman et al. 2020). According to Dai et al. (2014), GIA models provide better grounds for understanding the complexity of the conceptual model. Recently, in his work, Mahmoudi and Rasti-Barzoki (2018) explained the systematic process and steps to measure Deng's GIA model of two datasets represented as ω_0 and ω_η .

- Stage 1: As a first step to measure Deng's GIA, we present the equation of our two datasets δ_0 and δ_η , $\eta=1, 2, 3, \dots, \uparrow$, where:

$$\omega'_\eta = \frac{\omega\eta}{\omega\eta(1)} = (\omega'_\eta(1), \omega'_\eta(2), \dots, \omega'_\eta(n)); \eta = 0, 1, 2, 3, 4, \dots, m. \quad (1)$$

- Stage 2: This involves the calculation of the differences in the δ'_0 and δ'_η sequence, where $\eta=1, 2, 3, 4, \dots, \uparrow$ and is presented as

$$\Delta\eta(1) = \|\omega'_0(1) - \omega'_\eta(1)\|, \Delta = (\Delta\omega(1), \Delta\omega(2), \dots, \Delta\omega(n)), \eta = 1, 2, 3, \dots, m. \quad (2)$$

- Stage 3: We attempt to find the difference (maximum and minimum) between:

$$\mathbb{M} = \max_\eta \max_1 \Delta_\eta(\mathfrak{Z}), \quad (3)$$

$$\mathbf{m} = \max_\eta \max_1 \Delta_\eta(\mathfrak{Z}), \quad (4)$$

- Stage 4: Grey incidence coefficient is measured with the help of the following formula, where ζ is the distinctive coefficient and has the value of 0.5:
- Stage 5: Grey relational degree (GRG) is measured with Eq. (5)

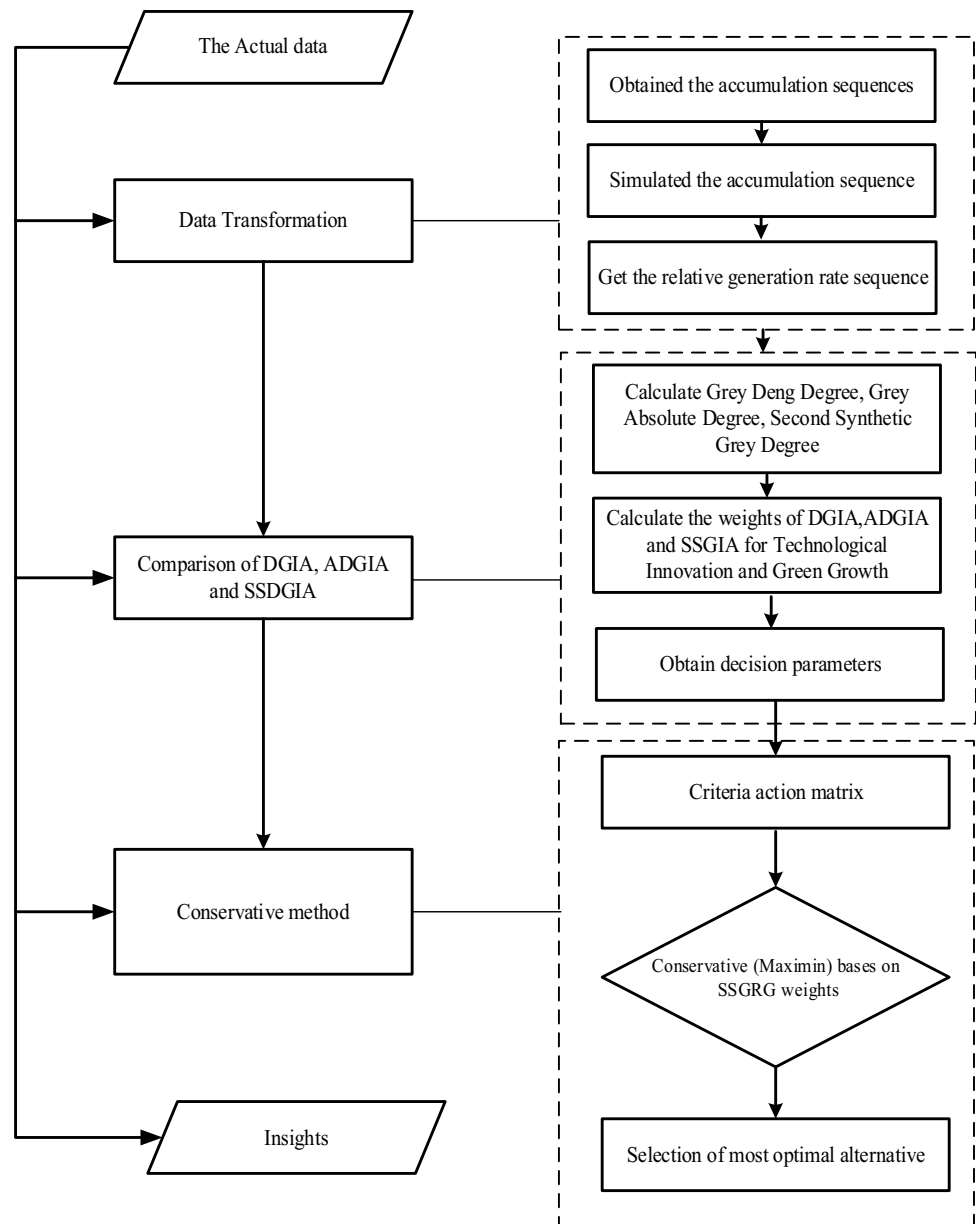
$$\xi_0\eta = \sum \mathfrak{Z} = 1n(\xi_0\eta(\mathfrak{Z}) \times \sqsubseteq 1); \eta = 1, 2, 3, 4, \dots, m. \quad (5)$$

In Eq. (5), $\sum \sqsubseteq 1=1$, and $\frac{1}{n}$ represents equally distributed criteria (Sarıkaya and Güllü 2015).

Absolute degree of GIA model

Deng's GIA model works better than traditional statistical methods to explain the relationship between study variables (Ikram et al. 2021c). It is known as the absolute GRA model or absolute degree of GIA (ADGIA) (J. Xu and Li 2018; Zhicai and Li 2018). Using this rigorous econometric technique in the investigation allows the researchers to gain

Fig. 5 Schematic framework of the study (source: Ikram et al. 2021b)



insights into the geometric proximity between the sequential study setup and the information on study parameters and their association with the designed framework (Irfan et al. 2021; Javed and Liu 2017). In order to analyze the GIA in the study model, it is necessary to access the two separate time periods ω_0 and ω_η .

- Stage 1: Use the different time intervals ω_0 and ω_η to measure the base point ω_0^0 and ω_1^0 .
- Stage 2: Estimate $|\gamma_0|$, $|\gamma_1|$, and $|\gamma_1 - \gamma_0|$.
- Stage 3: Calculate the ADGIA different time span (μ_{01}) using the δ_0 and δ_1 sequence. The formula used to estimate the ADGIA is given below:

$$\mu_{01} = \frac{1 + |\gamma_0| + |\gamma_1|}{1 + |\gamma_0| + |\gamma_1| + |\gamma_1 - \gamma_0|}. \quad (6)$$

Deng's degree of GIA

In the following stage of the analysis, the GRA model proposed by Deng suggests employing Deng's degree known as GRG (grey relational grade) or DGIA (degree of grey incidence analysis). The mathematical representation of DGIA is given below in Eqs. (7) and (8), where $\xi_{0\eta}(\mathfrak{T})$ is a grey relational coefficient (GRC).

$$\xi_{0\eta} = \sum_{i=1}^n (\xi_{0\eta}(\mathfrak{T}) \times \lambda_i); \eta = 1, 2, 3, 4 \dots, m, \quad (7)$$

Or

$$\xi_{0\eta} = \frac{1}{n} \sum_{i=1}^n (\xi_{0\eta}(\mathfrak{T})); \eta = 1, 2, 3, 4 \dots, m, \quad (8)$$

Equation (7) represents the weighted GRA/DGIA, and Eq. (8) shows the non-weighted scheme.

Definition 1: absolute degree of grey incidence analysis (ADGIA)

This study used the ADGIA model to estimate the weights of technological innovation, energy, environmental management systems, population growth, and green growth for Pakistan's case.

The formula for weight measurements is given below:

$$\beta_{01} = \frac{1 + |\gamma_0| + |\gamma_1|}{1 + |\gamma_0| + |\gamma_1| + |\gamma_1 - \gamma_0|} \quad (9)$$

Let the data sequence of system behavior be represented as $\omega_\eta = (\omega_\eta(1), \omega_\eta(2), \dots, \omega_\eta(n))$ and $\mathcal{C}$ indicate the sequence operator, which satisfies $\delta_\eta(L)\mathcal{C} = (\omega_\eta(L)\mathcal{C} - \omega_\eta(1); \mathfrak{T} = 1, 2, 3, \dots, n)$. Then, $\mathcal{C}$ is a point operator starting from zero and $\delta_i\mathcal{C}$, known as zero-scale point image of $\delta_i\mathcal{C}$ is often written as:

$$\omega_\eta\mathcal{C} = \omega_1^0 = (\omega_1^0(1), \omega_1^0(2), \dots, \omega_1^0(n)) \quad (10)$$

According to Liu et al. (Liu et al. 2017), we assume that the length of variables δ_i and δ_1 is the same as the one-time-interval sequences and δ_i and δ_1 are zero starting point images as follows:

$$\omega_\eta^0 = (\omega_\eta^0(1), \omega_\eta^0(2), \omega_\eta^0(3) \dots, \omega_\eta^0(n)), \quad (11)$$

$$\omega_\theta^0 = (\omega_\theta^0(1), \omega_\theta^0(2), \omega_\theta^0(3) \dots, \omega_\theta^0(n)). \quad (12)$$

Then

$$|\varsigma_\eta| = \left[\sum_{i=2}^{n-1} \omega_\eta^0(\mathfrak{T}) + \frac{1}{2} \omega_\eta^0(n) \right] \quad (13)$$

$$|\varsigma_\theta| = \left[\sum_{i=2}^{n-1} \omega_\theta^0(\mathfrak{T}) + \frac{1}{2} \omega_\theta^0(n) \right], \quad (14)$$

$$|\varsigma_\eta - \varsigma_\theta| = \left[\sum_{i=2}^{n-1} (\omega_\theta^0(\mathfrak{T}) - \omega_\eta^0(\mathfrak{T})) + \frac{1}{2} (\omega_\eta^0(n) - \omega_\theta^0(n)) \right], |\varsigma_\theta| = \left[\sum_{i=2}^{n-1} \omega_\theta^0(\mathfrak{T}) + \frac{1}{2} \omega_\theta^0(n) \right], \quad (15)$$

SSDGIA model (second synthetic degree of grey incidence analysis)

The SSGIA model is the next stage to calculate the proximity between two curves using the two data sequences mentioned earlier. This model presents the whole proximity of the factor series. SSDGIA is measured by incorporating the integral and partial points of the proximity between the curves. This model is proposed by Javed and Liu (2018) and estimated with the help of the following equation:

$$\beta_{\eta\theta} = \delta\epsilon_{\eta\theta} + (1 - \delta)\varpi_{\eta\theta}; \delta\epsilon[0,1], \quad (16)$$

$\beta_{\eta\theta}$ = SS degree of grey relation (SSGRG).

$\epsilon_{\eta\theta}$ = Absolute degree of grey relation (absolute GRG).

$\varpi_{\eta\theta}$ = Deng's degree of grey incidence/grey relation (GRG).

The basic principle behind GRA models is to utilize the degree of proximity for a geometric curve in an open data structure to evaluate for associations (Liu et al. 2017, p.68). There are three types of proximities in GRA models. If Deng's GRA model relates partial closeness/proximity, then absolute GRA relates integral closeness/proximity between two data sequences, while the second synthetic GRA aims to reveal a more comprehensive closeness/inclusive proximity (Ikram et al. 2021; Javed et al. 2019). The SSDGIA model was developed based on the SDGIA model's working standard. It is recommended that $\theta=0.5$ be considered when making decisions with a view to achieving a thorough positioning that equally combines the advantages of the two terms, i.e., γ and ϵ , without preferring one over the other, and if appropriate, θ could be changed. If someone wants to endorse γ , then the calculation of θ might be decreased at that point, but if ϵ is desired, then the total value can also be increased at that point.

Results and discussion

This study used GRA models to analyze the relationship between technological innovation, EMS, energy consumption, population growth, and green growth in Pakistan. The efficiency of GRA models makes them a better technique as they address endogeneity issues and grey values, thus providing an optimal solution (Badri Ahmadi et al. 2017). The analysis results for GRA models, Deng GRG, SSGRG, and absolute GRG are set forth in Tables 2, 3, 4, 5, 6. GRA models help us evaluate the association between technological innovation (R&D, patent application), EMS (ISO 14001), energy consumption, population growth, and green growth (CO₂

Table 2 Grey relational assessment for research and development

Country	Absolute GRG	Deng GRG	SSGRG
GHG emission from energy	0.5308	0.6791	0.6049
Renewable energy consumption	0.8394	0.6882	0.7092
Mortality from exposure to PM 2.5	0.9112	0.7963	0.8537
CO ₂ emissions	0.9791	0.7330	0.8561

Table 3 Grey relational assessment for patent applications

Country	Absolute GRG	Deng GRG	SSGRG
GHG emission from energy	0.5238	0.6814	0.6026
Renewable energy consumption	0.6851	0.6992	0.6921
Mortality from exposure to PM 2.5	0.8179	0.7312	0.77462
CO ₂ emissions	0.8705	0.7322	0.8013

Table 4 Grey relational assessment for energy consumption

Indicators	Absolute GRG	Deng GRG	SSGRG
GHG emission from energy	0.5436	0.4455	0.4945
Renewable energy consumption	0.5391	0.5455	0.5923
Mortality from exposure to PM 2.5	0.9292	0.6823	0.7658
CO ₂ emissions	0.8683	0.6959	0.7821

Table 5 Grey relational assessment for EMS

Indicators	Absolute GRG	Deng GRG	SSGRG
GHG emission from energy	0.6143	0.9424	0.7797
Renewable energy consumption	0.5952	0.8975	0.7523
Mortality from exposure to PM 2.5	0.6307	0.6392	0.6850
CO ₂ emissions	0.6267	0.9788	0.7824

Table 6 Grey relational assessment for population

Indicators	Absolute GRG	Deng GRG	SSGRG
GHG emission from energy	0.5201	0.5003	0.5602
Renewable energy consumption	0.6868	0.8204	0.7036
Mortality from exposure to PM 2.5	0.5158	0.5866	0.6012
CO ₂ emissions	0.6981	0.7601	0.7241

emissions, renewable energy, GHGs, PM 2.5) in Pakistan for the period of 2000–2019. The strength of this relationship is measured using a scale of 0 to 1, where 0 represents the weakest values and 1 indicates the most substantial relationship for GRG and SSGRG models. The Deng GRG model's scale has a standardized value of 0.5 to 1, where 0.5 indicates a weak relationship and 1 shows a strong relationship between study variables (Ikram et al. 2020; Javed et al. 2019). The assessment of technological innovation (R&D, patent application), EMS (ISO 14001), energy consumption, population growth, and green growth (CO₂ emissions, renewable energy, GHGs, PM 2.5) with respect to GRA models is presented in Fig. 6.

Table 2 shows the relationship between technological innovation (R&D) and green growth (CO₂ emissions, renewable energy, GHGs, PM 2.5) in Pakistan. Absolute GRG results show that CO₂ emissions have a substantial relationship with green growth in Pakistan with a value of 0.9791 followed by PM 2.5 (0.9112), which represents the most dangerous air pollutant for human health. Deng GRG results show that PM 2.5 got the highest-ranking value (0.7963), followed by CO₂ emissions (0.7330). PM 2.5 is responsible for severe lung and respiratory diseases and deaths (Diao et al. 2019). According to the WHO, every year, 40 million cases and 28,000 deaths due to severe lung and respiratory diseases are reported in Pakistan (Environmental health 2020). Results of R&D as an independent variable in Table 1 indicate the significant relationship between CO₂ and PM 2.5 emissions, which means that government spending on R&D will help Pakistan address CO₂ and PM 2.5 problems nationwide.

The SSGRG model findings rate CO₂ emission in first position (0.8561) with R&D and PM 2.5 (0.8537) in second position. The third position is occupied by renewable energy consumption with a value of 0.7092. Pakistan relies on fossil fuels to meet most of its energy and other economic demands. Although the country has launched various initiatives to support renewable energy production, most of Pakistan's future energy projects are still based on fossil fuels (Abid et al. 2020).

GRA assessment for R&D shows that GHG emissions from energy consumption in Pakistan do not share any significant relationship with R&D funding. Our results for PM 2.5 and CO₂ emissions are in line with the findings of Di Vita (2017) and Zhang et al. (2019), whose research revealed that technological innovation (R&D funding) helps lower CO₂ emissions and air pollution (PM 2.5) in a country. R&D funding in Pakistan is insufficient (Global economy, 2021) to boost its technological innovation, so the government should focus on these aspects to protect people's lives or aim to comply with international environmental laws.

Patent applications are a good proxy for measuring technological innovation in a country (Aghion et al. 2005; Berghäll 2016). Table 3 shows the relationship between

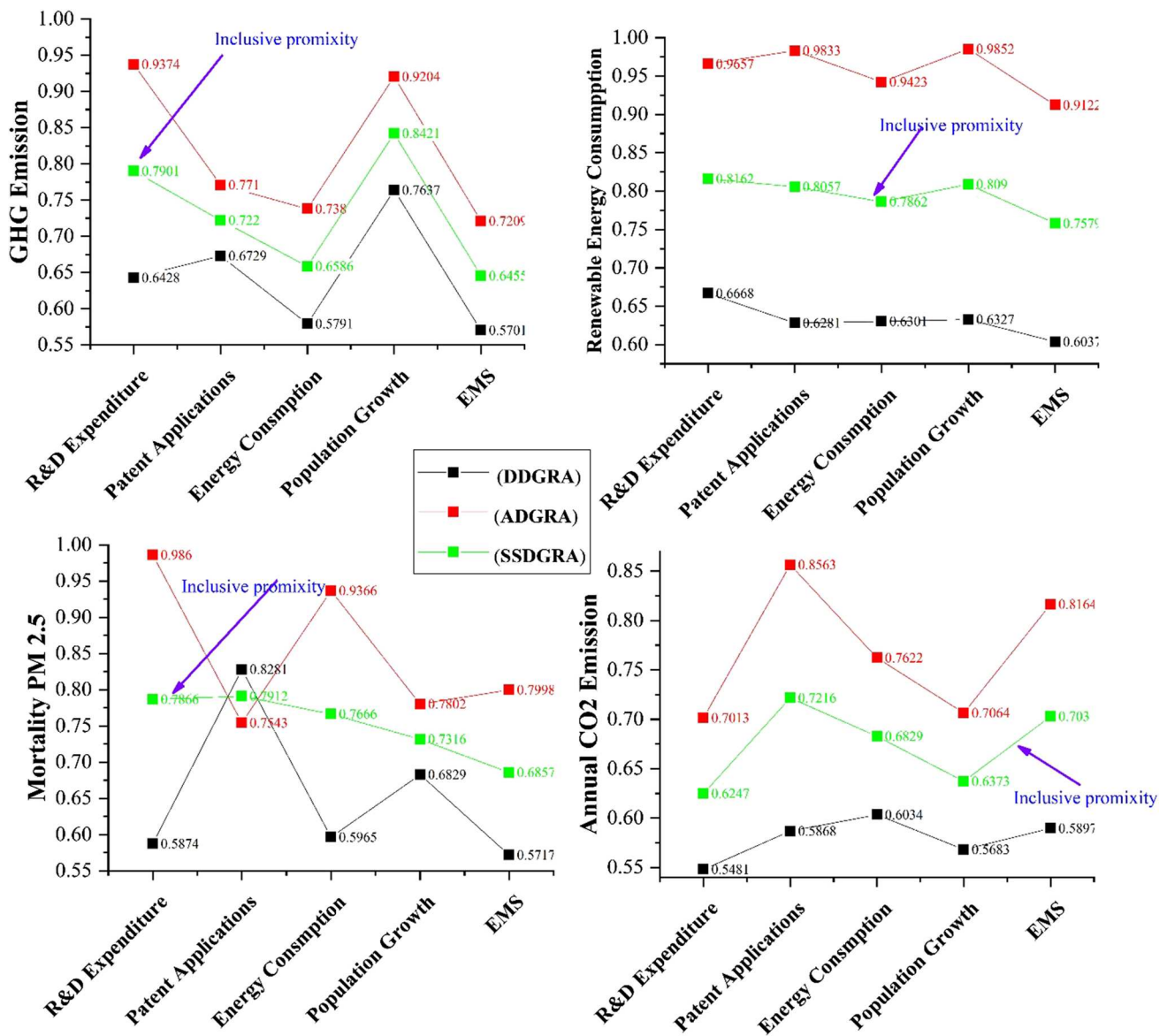


Fig. 6 GRA-based evaluation of technological innovation and green growth

patent applications (independent variable) and CO₂ emissions, renewable energy, GHG emissions, and PM 2.5 (dependent variables). The result of absolute GRG shows that CO₂ emissions with a weight value of 0.8705 share a significant relationship with patent applications in Pakistan, followed by PM (0.8179). Deng GRG and SSGRG models show similar results as absolute GRG and rank CO₂ emissions (0.7322 and 0.8013) and PM 2.5 (0.7312 and 0.7746) as factors that directly influence patent applications in the country. These findings are similar to the results of Geng (2016). These researchers found that patent applications are crucial factors that help lower CO₂ and other harmful emissions. There is a negative relationship between technological innovation and CO₂ emission, which postulates

that technological innovation can lead to a decrease in CO₂ emissions for Pakistan (Abid et al. 2020). However, as for Pakistan, renewable energy and GHG emissions share a significant relationship with patent applications. The number of patent applications is not progressing compared to other developing countries in Asia (The world bank 2018). There is a need to nourish technological innovation in Pakistan to achieve environmental sustainability.

GRA results for energy consumption with GHG emissions, renewable energy, CO₂ emissions, and PM 2.5 are presented in Table 4. Surprisingly, absolute GRG ranks PM 2.5 as the indicator that is most affected (0.9292) by energy consumption in Pakistan. According to a World Bank report, 30 million people in Pakistan suffer from severe respiratory

conditions every year, leading to 28,000 deaths (World Health Organization 2020). PM 2.5 is the most harmful air pollutant responsible for lung and chronic reparatory diseases (Diao et al. 2019). Pakistan produces most of its energy from fossil fuels (Wakeel et al. 2016), and our findings indicate that energy consumption is a significant cause of the rising level of PM 2.5. CO₂ emissions (0.8683) rank second among the various factors that are most affected by energy consumption. However, the results of absolute GRG, Deng GRG, and SSGRG suggest that factories' energy consumption does not share a significant relationship with GHG emissions since more than 70% of GHG emissions is caused by agriculture in Pakistan (Hussain et al. 2019). Deng GRG and SSGRG support the results of absolute GRG that PM 2.5 (0.6823 and 0.7658) and CO₂ emissions (0.7821 and 0.6959) are the main outcomes of energy consumption in Pakistan.

When we use EMS as an independent variable, the results of Deng GRG models show that EMS is strongly associated with CO₂ emissions with a weight of 0.9788. The second variable is GHG emissions (0.9424), which indicates the effectiveness of EMS in addressing environmental issues in Pakistan. Renewable energy also shares a significant relationship with EMS (0.8975), and PM 2.5 has a comparably weak relationship with EMS (0.6392) in Pakistan. The details are given in Table 5. The results of SSRG place CO₂ emissions first, followed by GHG emissions and renewable energy consumption. Our findings for EMS affecting renewable energy are similar to the results of Bellakhal et al. (2019) and Frondel et al. (2008). The absolute GRG shows that EMS effectively lowers air pollutants, and the PM 2.5 value is 0.6307. However, Deng GRG and SSGRG models show that EMS (ISO 14001) shares a more significant relationship with CO₂ and GHG emissions. These outcomes align with the findings of Muhammad et al. (2020), who also show in their analysis that ISO 14001 lowers CO₂ and GHG emissions. Organizations in Pakistan tend to employ EMS (ISO 14001) to improve environmental performance, but the number of organizations adopting ISO certifications is not promising (Muhammad Ikram et al. 2020).

Table 6 presents the relationship between population growth and GHG emissions, renewable energy, PM 2.5, and CO₂ emissions in Pakistan. Absolute GRG, Deng GRG, and SSGRG indicate that Pakistan's population growth is somehow responsible for CO₂ emissions as humans are natural emitters of CO₂ (Naqvi and Rehm 2014). The positive association between CO₂ emissions and population growth is backed by the results of Naqvi and Rehm (2014), who found a positive correlation between population and CO₂ emissions (Abid et al. 2020).

Population growth does not share a substantial relationship with GHGs and PM 2.5 in Pakistan. The second significant factor that is related to population growth is

renewable energy consumption in Pakistan for all three GRA models. The reason behind this factor is the awareness regarding environmental protection among the people. The study of Vo and Vo (2021) also verified this result and asserted that the population needs energy; as the population grows, so does its energy consumption. With rising awareness of the consequences of fossil fuel energy consumption, people tend to demand energy produced through clean energy sources. The relationship between renewable energy and population is also explained by Avtar et al. (2019), who stated that the population exerts pressure on domestic authorities to use alternative energy sources in the country. Therefore, a change in population is also subject to a change in the proportion of energy consumption from renewable energy sources.

$P=5$, $n=8$, and outcome = $v(s_i, D_j)$, where $i=1, 2, 3$ and $j=1, 2, 3, 4, 5 \dots 8$. s_1, s_2, s_3 , and s_4 show the GRA of technological innovation, energy, environmental management systems, and population growth with green growth in Pakistan. The SSDGRA (second synthetic degree of the grey relation-based matrix) is presented in Table 7, which unfolds the relationship between the study factors to arrange the variables and conclude the criteria response. Then, a decision-making model is applied to evaluate the impact of technological innovation, energy, environmental management systems, and population growth on green growth in Pakistan, and it is presented in Table 8.

Finally, we applied the conservative (maximin) model (Ikram et al. 2019) to obtain the optimal solution among all variables for green growth development in Pakistan based on the SSGRG values presented in Table 9. The conservative model shows the following optimal solution:

$$\max_{L_i} \{ \min_{A_k} v(A_p, C_k) \} = \max_{A_p} \left\{ \begin{array}{l} 0.4945 \\ 0.6921 \\ 0.6012 \\ 0.6241 \end{array} \right\} = 0.6921 (\text{Patent application})$$

This paper has investigated Pakistan's technological innovation, energy, EMS, population, and economic growth in a multi-framework. The study has separately analyzed the potential influence of individual independent variables on dependent variables. Figure 7 shows that patent applications, among all variables, have a significant impact on green growth in Pakistan. Over the years, patent applications have been used as drivers for economic growth. Patent applications can drive sustainability in a country and help abate the environmental crisis (Nitta 2005). Patent applications are a key indicator of technological innovation (Aghion et al. 2005) and help promote renewable energy technology and consumption (Popp 2012), which lower GHGs and other harmful emissions (Johnstone et al. 2010).

Table 7 Grey assessment ranking of technological innovation, EMS, energy, population, and green growth

Indicators	Grey relational model	Ranking
Research and development expenditure	Absolute GRG	CO ₂ emissions > mortality from exposure to PM 2.5 > renewable energy consumption > GHG emission
	Deng's GRG	Mortality from exposure to PM 2.5 > CO ₂ emission > renewable energy consumption > GHG emission
	SS GRG	CO ₂ emissions > voice and accountability > mortality from exposure to PM 2.5 > renewable energy consumption > GHG emission
Patent applications	Absolute GRG	CO ₂ emissions > mortality from exposure to PM 2.5 > renewable energy consumption > GHG emission
	Deng's GRG	CO ₂ emissions > mortality from exposure to PM 2.5 > renewable energy consumption > GHG emission
	SS GRG	CO ₂ emissions > mortality from exposure to PM 2.5 > renewable energy consumption > GHG emission
Energy consumption	Absolute GRG	Mortality from exposure to PM 2.5 > CO ₂ emissions > renewable energy consumption > GHG emission
	Deng's GRG	CO ₂ emissions > mortality from exposure to PM 2.5 > Renewable energy consumption > GHG emission
	SS GRG	CO ₂ emissions > mortality from exposure to PM 2.5 > Renewable energy consumption > GHG emission
EMS	Absolute GRG	Mortality from exposure to PM 2.5 > CO ₂ emissions > GHG emission > renewable energy consumption
	Deng's GRG	CO ₂ emissions > GHG emission > renewable energy consumption > mortality from exposure to PM 2.5
	SS GRG	CO ₂ emissions > GHG emission > renewable energy consumption > mortality from exposure to PM 2.5
Population growth	Absolute GRG	CO ₂ emissions > renewable energy consumption > GHG emission > mortality from exposure to PM 2.5
	Deng's GRG	Renewable energy consumption > mortality from exposure to PM 2.5 > CO ₂ emissions > GHG emission
	SS GRG	CO ₂ emissions > renewable energy consumption > GHG emission > mortality from exposure to PM 2.5

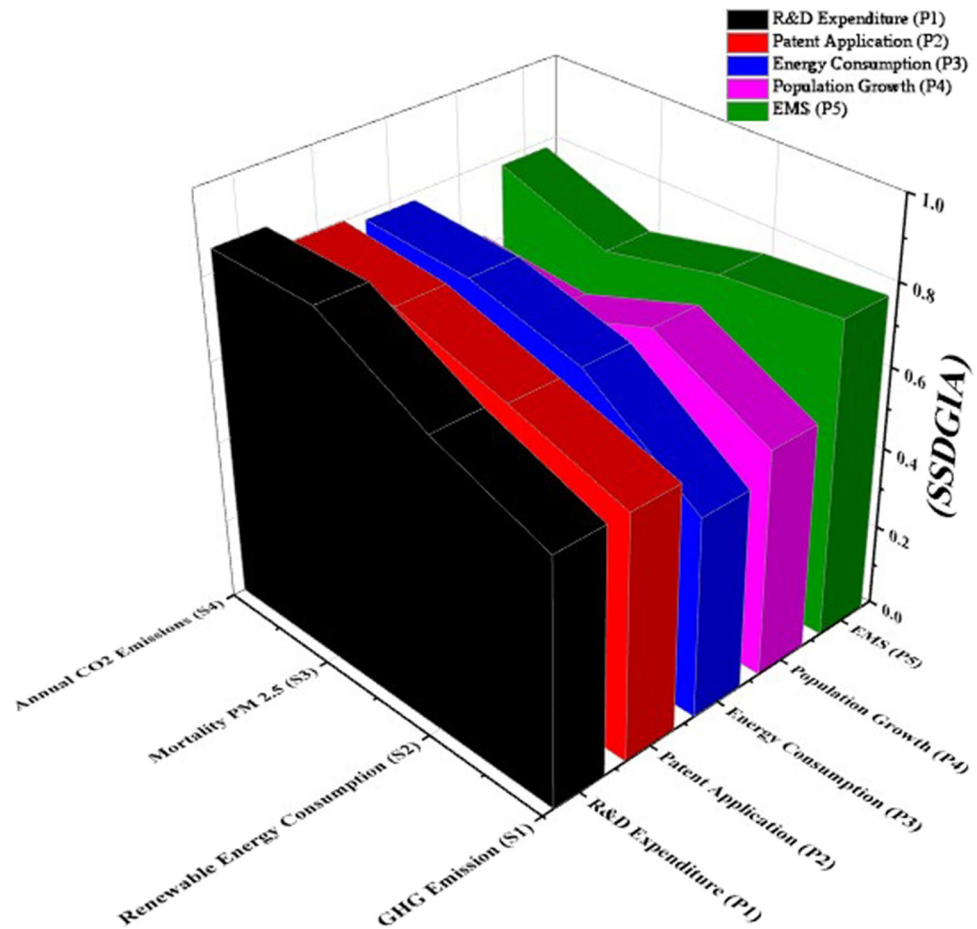
Table 8 Description of goal and decision parameters

Goal	Measuring grey relation (association) between technological innovation, population growth, renewable energy, environmental management systems, and green growth in Pakistan
State of nature/criteria (P_j), $P = 1, 2, \dots, k$	<i>Patent applications are positively related to green growth (P1)</i> <i>R&D is positively related to green growth (P2)</i> <i>EMSs are positively associated with green growth (P3)</i> <i>Energy consumption is negatively related to green growth (P4)</i> <i>Population growth is negatively related to green growth (P5)</i>
Alternative actions (S_i), $S = 1, 2, \dots, m$	Grey relation was superior in GHG emission ($S1$) Grey relation was superior in renewable energy consumption ($S2$) Grey relation was superior in mortality from exposure to PM 2.5 ($S3$) Grey relation was superior in annual CO ₂ emissions ($S4$)

Table 9 SSGRG criteria-based matrix

SSGRG	P ₁	P ₂	P ₃	P ₄	P ₅
S ₁	0.6049	0.6026	0.4945	0.5602	0.7797
S ₂	0.7092	0.6921	0.6923	0.7036	0.7523
S ₃	0.8537	0.77462	0.7658	0.6012	0.6850
S ₄	0.8561	0.8013	0.7821	0.6241	0.6440

Fig. 7 Conservative (maximin)–based evaluation of technological innovation and green growth



Conclusion and policy implications

Conclusion

Green growth is pervasive in attaining sustainable development in a country. This paper has explored the dynamic linkage between technological innovation, EMS (seen as a green form of organizational innovation), environmental challenges (namely, energy consumption and population growth), and green growth in the context of Pakistan by using a novel advanced mathematical GRA modeling approach.

Pakistan is a developing country facing acute energy and environmental crisis that needs to be addressed. Despite the excellent potential for renewables, Pakistan has been utilizing fossil fuels combined with traditional technologies to meet the country's economic and social demands. Our results indicate that technological innovation (patent applications and R&D) are core elements that help develop green growth in Pakistan. R&D and patent applications are efficient in lowering CO₂ emissions, PM 2.5, and GHG emissions and in promoting renewable energy consumption. Technological innovation helps a country transition from

traditional technology to green technology, thus improving environmental performance and boosting production, operation, and process efficiency.

The relationship between energy and green growth shows that energy consumption in factories is a significant cause of the high CO₂ emissions in Pakistan but not directly linked to other GHG emissions in the country. Pakistan is an agricultural country, and other GHG emissions (more than 70%) come from the agricultural sector (Hussain et al. 2019). Energy consumption is also responsible for PM 2.5, an air pollutant having a direct impact on human health. Energy consumption, however, does not exhibit any strong relationship with renewable energy in the country. Although the government has launched initiatives to promote renewable energy on an industrial scale, fossil fuels are still the top choice, which calls for a change in the whole system.

The government must support renewable energy sources and reduce taxes to help industries transition from conventional to alternative energy sources. When we examine the relationship between EMS and green growth, the results show a significant relationship between EMS, CO₂, and renewable energy. EMS helps reduce CO₂ and GHG emissions. EMS shares a significant positive relationship with

renewable energy, postulating policy implications for Pakistan to implement EMS in order to increase environmental performance. Population growth, however, does not have a meaningful impact on green growth. Despite the existence of an association between the variables, this relationship is not very strong. Results indicate that an increase in population actually causes an increase in CO₂ emissions in Pakistan. The second association was observed between population and renewable energy, which implies that in raising social awareness of environmental protection and preservation, people put pressure on the authorities to protect the environment.

Policy implications

Our study offers the following policy implications for stakeholders. First, the government's budget for R&D is not competitive when compared to neighboring developing countries. The Pakistani government must allocate a sufficient budget to R&D as this will help the country boost its technological innovation. The number of national patent applications is also significantly low (The World Bank 2018), which requires substantial efforts from the government and an effective policy to support locals who want to excel in the technology domain to improve and increase the number of patent applications. The procedure and budget will help local businessmen, technologists, and SMEs develop and employ local technology to produce energy from renewables. Technological innovation reduces CO₂ and other emissions and promotes renewable resources and technology to meet national economic demands.

Second, developing countries rely on fossil fuels to meet energy and production demands. It is not easy for a country like Pakistan to shift its industrial operations from fossil fuels to renewables. It requires massive investment in machinery and resource transition. Industries need support to change their operational functions or import technology or technicians in the early stages. The government of Pakistan must take steps to support industries to shift their operations from fossil fuels to renewables, as Pakistan has sufficient renewable potential. The government must attract foreign investors and technology leaders to invest in technological infrastructure and businesses in Pakistan. The goal cannot be achieved by politics alone; substantial efforts, plans, and programs are needed to show the world what this country has to offer, and imports and local taxes should be reduced to support FDI and local industries.

Third, environmental management systems support green growth progress, but the number of environmental management systems in Pakistan is not promising. State and provincial authorities need to raise awareness of the importance of EMSs within organizations. Pakistan has laws to protect the environment. The government should merge these laws with

awareness programs to show the need to adopt environmental management systems in order to protect the environment. Fourth, as a developing country with tight job market and competitive opportunities, it must control the population growth with the help of all levels of government hierarchy.

This study is not without limitations. Since the authors focused on macro-level state factors and variables to track green growth development, the effects of the micro-level sample covering provincial progress towards green growth and organizational level impact have therefore been ignored. Besides technological innovation and environmental challenges, other factors accelerate or hinder green growth development. Therefore, future researchers may consider exploring foreign direct investment, corporate social responsibility, and the institutional relationship with green growth in Pakistan by using advanced econometric techniques at the micro and macro levels. In multiple frameworks, it is hard to include all green growth indicators in one study, so future researchers may explore the dimensions of green growth at the organizational level with a detailed review of Pakistan's circumstances. Green growth is an opportunity to achieve prolonged sustainable development in economic, social, and environmental structures. Hence, an academic from a different field should explore the phenomenon to help the country move forward in a sustainable direction.

Author contribution Conceptualization, Nabila Abid and Muhammad Ikram; formal analysis, Muhammad Ikram; investigation, Nabila Abid, Federica Ceci, and Muhammad Ikram; methodology, Muhammad Ikram; project administration and supervision, Federica Ceci and Muhammad Ikram; validation, Nabila Abid, Federica Ceci, and Muhammad Ikram; writing—original draft, Nabila Abid; writing—review and editing, Nabila Abid, Federica Ceci, and Muhammad Ikram.

Availability of data and materials All data generated or analyzed during this study are available online.

Declarations

Ethics approval Not applicable

Consent to participate Not applicable

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

References

- Abid N, Ikram M, Wu J, Ferasso M (2021) Towards environmental sustainability: exploring the nexus among ISO 14001, governance indicators and green economy in Pakistan. *Sustain Prod Consum* 27:653–666. <https://doi.org/10.1016/j.spc.2021.01.024>
- Abid N, Wu J, Ahmad F, Draz MU, Chandio AA, Xu H (2020) Incorporating environmental pollution and human development in the

- energy-growth nexus: a novel long run investigation for Pakistan. *Int J Environ Res Public Health* 17(14):1–22. <https://doi.org/10.3390/ijerph17145154>
- Acosta LA, Maharjan P, Peyriere HM, Mamiit RJ (2020) Natural capital protection indicators: measuring performance in achieving the Sustainable Development Goals for green growth transition. *Environmental and Sustainability Indicators* 8(July):100069. <https://doi.org/10.1016/j.indic.2020.100069>
- Aghion P, Bloom N, Blundell R, Griffith R, Howitt P (2005) Competition and innovation: an inverted-U relationship. *Q J Econ* 120(2):701–728. <https://doi.org/10.1093/qje/120.2.701>
- Aghion P, Dechezleprêtre A, Hémous D, Martin R, van Reenen J (2016) Carbon taxes, path dependency, and directed technical change: evidence from the auto industry. *J Polit Econ* 124(1):1–51. <https://doi.org/10.1086/684581>
- Aghion P, Dechezleprêtre A, Hemous D, Martin R, Van Reenen JM (2013) Carbon taxes, path dependency and directed technical change: evidence from the auto industry. SSRN Electron J. <https://doi.org/10.2139/ssrn.2202047>
- Aguilera-Caracuel J, Ortiz-de-Mandojana N (2013) Green innovation and financial performance: an institutional approach. *Organization and Environment* 26(4). <https://doi.org/10.1177/1086026613507931>
- Ahmad M, Jiang P, Majeed A, Umar M, Khan Z, Muhammad S (2020) The dynamic impact of natural resources, technological innovations and economic growth on ecological footprint: an advanced panel data estimation. *Resour Policy* 69(July):101817. <https://doi.org/10.1016/j.resourpol.2020.101817>
- Alam MJ, Begum IA, Buysse J, Rahman S, Van Huylbroeck G (2011) Dynamic modeling of causal relationship between energy consumption, CO₂ emissions and economic growth in India. *Renew Sustain Energy Rev* 15(6):3243–3251. <https://doi.org/10.1016/j.rser.2011.04.029>
- Naqvi Ali Asjad, Rehm Miriam (2014) A multi-agent model of a low income economy: simulating the distributional effects of natural disasters. *J Econ Interact Coord* 9:275–309
- Alkaher I, Carmi N (2019) Is population growth an environmental problem? Teachers' perceptions and attitudes towards including it in their teaching. *Sustainability (Switzerland)*, 11(7). <https://doi.org/10.3390/su11071994>
- Apergis N, Payne JE (2009) CO₂ emissions, energy usage, and output in Central America. *Energy Policy* 37(8):3282–3286. <https://doi.org/10.1016/j.enpol.2009.03.048>
- Aragón-Correa JA, Rubio-López EA (2007) Proactive corporate environmental strategies: myths and misunderstandings. *Long Range Planning* 40(3):357–381. <https://doi.org/10.1016/j.lrp.2007.02.008>
- Aravind D (2012) Learning and innovation in the context of process-focused management practices: the case of an environmental management system. *J Eng Technol Manag* 29(3):415–433. <https://doi.org/10.1016/j.jengtecman.2012.05.001>
- Arce ME, Saavedra Á, Míguez JL, Granada E (2015) The use of grey-based methods in multi-criteria decision analysis for the evaluation of sustainable energy systems: a review. *Renew Sustain Energy Rev* 47:924–932. <https://doi.org/10.1016/j.rser.2015.03.010>
- Audretsch DB, Belitski M (2020) The role of R&D and knowledge spillovers in innovation and productivity. *Eur Econ Rev* 123:103391. <https://doi.org/10.1016/j.euroecorev.2020.103391>
- Avtar R, Tripathi S, Aggarwal AK (2019) Population – urbanization – energy nexus : a review. 1–21
- Badri Ahmadi H, Hashemi Petrudi SH, Wang X (2017) Integrating sustainability into supplier selection with analytical hierarchy process and improved grey relational analysis: a case of telecom industry. *Int J Adv Manuf Technol* 90(9–12):2413–2427. <https://doi.org/10.1007/s00170-016-9518-z>
- Bagheri M, Guevara Z, Alikarami M, Kennedy CA, Doluweera G (2018) Green growth planning: a multi-factor energy input-output analysis of the Canadian economy. *Energy Economics* 74:708–720. <https://doi.org/10.1016/j.eneco.2018.07.015>
- Bansal P, Hunter T (2003) Strategic explanations for the early adoption of ISO 14001. *J Bus Ethics* 46(3):289–299. <https://doi.org/10.1023/A:1025536731830>
- Barbier EB (2016) Building the green economy. *Can Public Policy* 42(s1):1–9. <https://doi.org/10.3138/cpp.2015-017>
- Bekun FV, Emir F, Sarkodie SA (2019) Another look at the relationship between energy consumption, carbon dioxide emissions, and economic growth in South Africa. *Sci Total Environ* 655:759–765. <https://doi.org/10.1016/j.scitotenv.2018.11.271>
- Bellakhal R, Ben Kheder S, Haffoudhi H (2019) Institutional and market factors driving renewable energy development in MENA region: a panel data approach. *Energy Economics*, 84(c). <https://doi.org/10.1016/j.eneco.2019.104541>
- Berghäll E (2016) Innovation, competition and technical efficiency. *Cogent Business and Management* 3(1):1–23. <https://doi.org/10.1080/23311975.2016.1199522>
- Bhattacharya M, Paramati SR, Ozturk I, Bhattacharya S (2016) The effect of renewable energy consumption on economic growth: evidence from top 38 countries. *Appl Energy* 162:733–741. <https://doi.org/10.1016/j.apenergy.2015.10.104>
- Boiral O, Henri JF (2012) Modelling the impact of ISO 14001 on environmental performance: a comparative approach. *J Environ Manage* 99:84–97. <https://doi.org/10.1016/j.jenvman.2012.01.007>
- Carrillo-Labela R, Fort F, Parras-Rosa M (2020) Motives, barriers, and expected benefits of ISO 14001 in the agri-food sector. *Sustainability (switzerland)* 12(5):1–17. <https://doi.org/10.3390/su12051724>
- Chen Y, Lee CC (2020) Does technological innovation reduce CO₂ emissions? Cross-Country Evidence. *J Clean Prod* 263:121550. <https://doi.org/10.1016/j.jclepro.2020.121550>
- Chiarini A (2017) Setting strategies outside a typical environmental perspective using ISO 14001 certification. *Bus Strateg Environ* 26(6):844–854. <https://doi.org/10.1002/bse.1969>
- Coccia M, Watts J (2020) A theory of the evolution of technology: technological parasitism and the implications for innovation management. *J Eng Technol Manag* 55(December 2019), 101552. <https://doi.org/10.1016/j.jengtecman.2019.11.003>
- Cuerva MC, Triguero-Cano Á, Córcoles D (2014) Drivers of green and non-green innovation: empirical evidence in low-tech SMEs. *J Clean Prod* 68:104–113. <https://doi.org/10.1016/j.jclepro.2013.10.049>
- Cui H, Zhao T, Wu R (2018) CO₂ emissions from China's power industry: policy implications from both macro and micro perspectives. *J Clean Prod* 200:746–755. <https://doi.org/10.1016/j.jclepro.2018.07.330>
- Curkovic S, Sroufe R (2011) Using ISO 14001 to promote a sustainable supply chain strategy. *Bus Strateg Environ* 20(2):71–93. <https://doi.org/10.1002/bse.671>
- da Fonseca LMCM (2015) ISO 14001:2015: an improved tool for sustainability. *J Ind Eng Manag* 8(1):35–50. <https://doi.org/10.3926/jiem.1298>
- Dai J, Liu X, Hu F (2014) Research and application for grey relational analysis in multigranularity based on normality grey number. *Sci World J* 2014. <https://doi.org/10.1155/2014/312645>
- Danish, & Ulucak, R. (2020) How do environmental technologies affect green growth? Evidence from BRICS economies. *Sci Total Environ* 712:136504. <https://doi.org/10.1016/j.scitotenv.2020.136504>
- Darnall N (2006) Why firms mandate ISO 14001 certification. *Bus Soc* 45:354–381. <https://doi.org/10.1177/0007650306289387>
- Darnall N, Henriques I, Sadosky P (2008) Do environmental management systems improve business performance in an international setting? *J Int Manag* 14(4):364–376. <https://doi.org/10.1016/j.intman.2007.09.006>
- Davis M, Ahiduzzaman M, Kumar A (2018) How will Canada's greenhouse gas emissions change by 2050? A disaggregated analysis of past and future greenhouse gas emissions using bottom-up energy modelling and Sankey diagrams. *Appl*

- Energy 220(April):754–786. <https://doi.org/10.1016/j.apenergy.2018.03.064>
- Di Vita G (2017) The international strengthening of IPR and air pollution abatement: the role of the TRIPs agreement. *Technol Econ Dev Econ* 23(2):316–338. <https://doi.org/10.3846/20294913.2015.1072752>
- Diao M, Holloway T, Choi S, O'Neill SM, Al-Hamdan MZ, Van Donkelaar A, Martin RV, Jin X, Fiore AM, Henze DK, Lacey F, Kinney PL, Freedman F, Larkin NK, Zou Y, Kelly JT, Vaidyanathan A (2019) Methods, availability, and applications of PM_{2.5} exposure estimates derived from ground measurements, satellite, and atmospheric models. *J Air Waste Manag Assoc* 69(12):1391–1414. <https://doi.org/10.1080/10962247.2019.1668498>
- Du J, Zhang J, Li X (2020) What is the mechanism of resource dependence and high-quality economic development? An empirical test from China. *Sustainability (Switzerland)* 12(19). <https://doi.org/10.3390/su12198144>
- D'Souza C, Marjoribanks T, Young S, Mort GS, MN, JJJ (2019) Environmental management systems: an alternative marketing strategy for sustainability. *J Strateg Mark* 27(5):417–434. <https://doi.org/10.1080/0965254X.2018.1430054>
- Feng Y, Ning M, Lei Y, Sun Y, Liu W, Wang J (2019) Defending blue sky in China: effectiveness of the “air pollution prevention and control action plan” on air quality improvements from 2013 to 2017. *J Environ Manage* 252(August):109603. <https://doi.org/10.1016/j.jenvman.2019.109603>
- Fernández Y, Fernández López MA, Olmedillas Blanco B (2018) Innovation for sustainability: the impact of R&D spending on CO₂ emissions. *J Clean Prod* 172:3459–3467. <https://doi.org/10.1016/j.jclepro.2017.11.001>
- Fisher-Vanden K, Wing IS (2008) Accounting for quality: Issues with modeling the impact of R&D on economic growth and carbon emissions in developing economies. *Energy Economics* 30(6):2771–2784
- Fronzel M, Horbach J, Rennings K (2008) What triggers environmental management and innovation? Empirical evidence for Germany. *Ecol Econ* 66(1):153–160. <https://doi.org/10.1016/j.ecolecon.2007.08.016>
- Geng J (2016) Technological innovation and renewable energy development: evidence based on patent counts. Jiang-Bo Geng and Qiang Ji *. *International Journal of Global Environmental Issues* 15(3). <https://doi.org/10.1504/IJGENVI.2016.076945>
- Hamit-Haggar M (2012) Greenhouse gas emissions, energy consumption and economic growth: a panel cointegration analysis from Canadian industrial sector perspective. *Energy Economics* 34(1):358–364. <https://doi.org/10.1016/j.eneco.2011.06.005>
- Hao LN, Umar M, Khan Z, Ali W (2021) Green growth and low carbon emission in G7 countries: how critical the network of environmental taxes, renewable energy and human capital is? *Sci Total Environ* 752:141853. <https://doi.org/10.1016/j.scitotenv.2020.141853>
- Henriques I, Sadorsky P (2007) Environmental technical and administrative innovations in the Canadian manufacturing industry. *Bus Strateg Environ* 16(2):119–132. <https://doi.org/10.1002/bse.475>
- Hussain M, Butt AR, Uzma F, Ahmed R, Islam T, Yousaf B (2019) A comprehensive review of sectorial contribution towards greenhouse gas emissions and progress in carbon capture and storage in Pakistan. *Greenhouse Gases: Science and Technology* 9(4):617–636. <https://doi.org/10.1002/ghg.1890>
- Ikram M, Zhou P, Shah SAA, Liu GQ (2019) Do environmental management systems help improve corporate sustainable development? Evidence from manufacturing companies in Pakistan. *J Clean Prod* 226:628–641. <https://doi.org/10.1016/j.jclepro.2019.03.265>
- Ikram M, Zhang Q, Sroufe R, Shah SZA (2020) Towards a sustainable environment: the nexus between ISO 14001, renewable energy consumption, access to electricity, agriculture and CO₂ emissions in SAARC countries. *Sustain Prod Consum* 22:218–230. <https://doi.org/10.1016/j.spc.2020.03.011>
- Ikram M, Shen Y, Ferasso M, D'Adamo I (2021a) Intensifying effects of COVID-19 on economic growth, logistics performance, environmental sustainability and quality management: evidence from Asian countries. *J Asia Bus Stud*. <https://doi.org/10.1108/JABS-07-2021-0316>
- Ikram M, Zhang Q, Sroufe R, Ferasso M (2021b) Contribution of certification bodies and sustainability standards to sustainable development goals: an integrated grey systems approach. *Sustain Prod Consum* 28:326–345. <https://doi.org/10.1016/j.spc.2021.05.019>
- Ikram M, Ferasso M, Sroufe R, Zhang Q (2021c) Assessing green technology indicators for cleaner production and sustainable investments in a developing country context. *J Clean Prod* 322:129090
- International Energy Agency (IEA) (2018) Global energy & CO₂ status report: the latest trends in energy and emissions in 2018. <https://www.iea.org/geco/>
- Irfan M, Ikram M, Ahmad M et al (2021) Does temperature matter for COVID-19 transmissibility? Evidence across Pakistani provinces. *Environ Sci Pollut Res* 28:59705–59719. <https://doi.org/10.1007/s11356-021-14875-6>
- Javed SA, Mahmoudi A, Khan A (2018) Investigation of drilling parameters on hybrid polymer composites using grey relational analysis, regression, fuzzy logic, and ANN models: a critical note. *Journal of the Brazilian Society of Mechanical Sciences and Engineering*, 40(5). <https://doi.org/10.1007/s40430-018-1470-4>
- Javed SA, Khan AM, Dong W, Raza A, Liu S (2019) Systems evaluation through new grey relational analysis approach: an application on thermal conductivity-petrophysical parameters' relationships. *Processes* 7(6):1–19. <https://doi.org/10.3390/pr7060348>
- Javed SA, Liu S (2018) Evaluation of outpatient satisfaction and service quality of Pakistani healthcare projects: application of a novel synthetic grey incidence analysis model. *Grey Systems: Theory and Application* 8(4):462–480
- Javed SA, Liu S (2017) Evaluation of project management knowledge areas using grey incidence model and AHP. *International Conference on Grey Systems and Intelligent Services (GSIS) 2017*:120–120. <https://doi.org/10.1109/GSIS.2017.8077684>
- Johnstone N, Haščič I, Popp D (2010) Renewable energy policies and technological innovation: evidence based on patent counts. *Environ Resource Econ* 45(1):133–155. <https://doi.org/10.1007/s10640-009-9309-1>
- Ju-Long D (1982) Control problems of grey systems. *Syst Control Lett* 1(5):288–294. [https://doi.org/10.1016/S0167-6911\(82\)80025-X](https://doi.org/10.1016/S0167-6911(82)80025-X)
- Jung S, An KJ, Dodbiba G, Fujita T (2012) Regional energy-related carbon emission characteristics and potential mitigation in eco-industrial parks in South Korea: logarithmic mean Divisia index analysis based on the Kaya identity. *Energy* 46(1):231–241. <https://doi.org/10.1016/j.energy.2012.08.028>
- Kammerer D (2009) The effects of customer benefit and regulation on environmental product innovation. Empirical evidence from appliance manufacturers in Germany. *Ecol Econ* 68(8–9):2285–2295. <https://doi.org/10.1016/j.ecolecon.2009.02.016>
- Kang SJ (2015) Green growth and sustainable development in G20: performance and prospects. *The International Monetary System, Energy and Sustainable Development* 138:273–293. <https://doi.org/10.4324/9781315732275>
- Kantor I, Fowler MW, Hajimiragha A, Elkamel A (2010) Air quality and environmental impacts of alternative vehicle technologies in Ontario, Canada. *Int J Hydrog Energy* 35(10):5145–5153. <https://doi.org/10.1016/j.ijhydene.2009.08.071>
- Köne AI, Büke T (2010) Forecasting of CO₂ emissions from fuel combustion using trend analysis. In *Renewable and Sustainable Energy Reviews* (Vol. 14, Issue 9, pp. 2906–2915). <https://doi.org/10.1016/j.rser.2010.06.006>
- Könnölä T, Unruh GC (2007) Really changing the course: the limitations of environmental management systems for innovation. *Bus Strategy Environ* 16(8):525–537

- Kunapatarawong R, Martínez-Ros E (2016) Towards green growth: how does green innovation affect employment? *Res Policy* 45(6):1218–1232. <https://doi.org/10.1016/j.respol.2016.03.013>
- Laskurain I, Ibarloza A, Larrea A, Allur E (2017) Contribution to energy management of the main standards for environmental management systems: the case of ISO 14001 and EMAS. *Energies*, 10(11). <https://doi.org/10.3390/en10111758>
- Latif S, Shahid M, Mahmood H, Shahbaz M (2014) Coal consumption: an alternate energy resource to fuel economic growth in Pakistan. 36:282–287. <https://doi.org/10.1016/j.econmod.2013.09.046>
- Li X, Du J, Long H (2019a) Green development behavior and performance of industrial enterprises based on grounded theory study: evidence from China. *Sustainability (switzerland)* 11(15):1–19. <https://doi.org/10.3390/su11154133>
- Li X, Du J, Long H (2019b) Theoretical framework and formation mechanism of the green development system model in China. *Environmental Development* 32(October):100465. <https://doi.org/10.1016/j.envdev.2019.100465>
- Li X, Du J, Long H (2020) Mechanism for green development behavior and performance of industrial enterprises (GDBP-IE) using partial least squares structural equation modeling (PLS-SEM). *Int J Environ Res Public Health* 17(22):1–19. <https://doi.org/10.3390/ijerph17228450>
- Li X, Long H (2020) Research focus, frontier and knowledge base of green technology in China: metrological research based on mapping knowledge domains. *Polish Journal of Environmental Studies* 29(5):3003–3011. <https://doi.org/10.15244/pjoes/114500>
- Liu S, Zhang H, Yang Y (2017) Explanation of terms of grey incidence analysis models. *Grey Systems: Theory and Application* 7(1):136–142
- Mahmoudi R, Rasti-Barzoki M (2018) Sustainable supply chains under government intervention with a real-world case study: An evolutionary game theoretic approach. *Comput Ind Eng* 116:130–143
- Mazzi A, Toniolo S, Catto S, De Lorenzi V, Scipioni A (2017) The combination of an environmental management system and life cycle assessment at the territorial level. *Environ Impact Assess Rev* 63:59–71. <https://doi.org/10.1016/j.eiar.2016.11.004>
- Mensah CN, Long X, Boamah KB, Bediako IA, Dauda L, Salman M (2018) The effect of innovation on CO2 emissions of OECD countries. *Environ Sci Pollut Res* 25(29):29678–29698. <https://doi.org/10.1007/s11356-018-2968-0>
- Merino-Saum A, Clement J, Wyss R, Baldi MG (2020) Unpacking the green economy concept: a quantitative analysis of 140 definitions. *J Clean Prod* 242:118339. <https://doi.org/10.1016/j.jclepro.2019.118339>
- Miao L (2017) Examining the impact factors of urban residential energy consumption and CO2 emissions in China – evidence from city-level data. *Ecol Ind* 73:29–37. <https://doi.org/10.1016/j.ecolind.2016.09.031>
- Mungai EM, Ndiritu SW, Rajwani T (2020) Do voluntary environmental management systems improve environmental performance? Evidence from waste management by Kenyan firms. *J Clean Prod* 265:121636. <https://doi.org/10.1016/j.jclepro.2020.121636>
- Naveh E, Meilich O, Marcus A (2006) The effects of administrative innovation implementation on performance: an organizational learning approach. *Strateg Organ* 4(3):275–302. <https://doi.org/10.1177/14761270060066969>
- Nitta I (2005) Proposal for a green patent system: implications for sustainable development and climate change. *Sustain Dev Law Policy* 5(2):61–65
- Odeh NA, Cockerill TT (2008) Life cycle GHG assessment of fossil fuel power plants with carbon capture and storage. *Energy Policy* 36(1):367–380. <https://doi.org/10.1016/j.enpol.2007.09.026>
- OECD Green Growth Studies (2011) Towards green growth: a summary for policy maker. <https://www.oecd.org/env/towards-green-growth-9789264111318-en.htm>
- Osobajo OA, Otitoju A, Otitoju MA, Oke A (2020) The impact of energy consumption and economic growth on carbon dioxide emissions. *Sustainability (switzerland)* 12(19):1–16. <https://doi.org/10.3390/SU12197965>
- Popp D (2012) The role of technological change in green growth. World Bank Policy Research Working Paper, 6239. <https://doi.org/10.3386/w18506>
- Raines SS (2002) Implementing ISO 14001 - an international survey assessing the benefits of certification. *Corp Environ Strateg* 9(4):418–426. [https://doi.org/10.1016/S1066-7938\(02\)00009-X](https://doi.org/10.1016/S1066-7938(02)00009-X)
- Ramanujam R, Muthukrishnan N, Raju R (2011) Optimization of cutting parameters for turning Al-SiC(10p) MMC using ANOVA and grey relational analysis. *Int J Precis Eng Manuf* 12(4):651–656. <https://doi.org/10.1007/s12541-011-0084-x>
- Rehman E, Ikram M, Feng MT, Rehman S (2020) Sectoral-based CO2 emissions of Pakistan: a novel grey relation analysis (GRA) approach. *Environ Sci Pollut Res* 27(23):29118–29129. <https://doi.org/10.1007/s11356-020-09237-7>
- Roberts G, Wooster MJ (2021) Global impact of landscape fire emissions on surface level PM2.5 concentrations, air quality exposure and population mortality. *Atmospheric Environment* 118210. <https://doi.org/10.1016/j.atmosenv.2021.118210>
- Rosa EA, Dietz T (2012) Human drivers of national greenhouse-gas emissions. *Nat Clim Chang* 2(8):581–586. <https://doi.org/10.1038/nclimate1506>
- Ruggieri M, Plaia A (2012) An aggregate AQI: comparing different standardizations and introducing a variability index. *Sci Total Environ* 420:263–272. <https://doi.org/10.1016/j.scitotenv.2011.09.019>
- Rusko M, Sablik J, Marková P, Lach M, Friedrich S (2014) Sustainable development, quality management system and environmental management system in Slovak Republic. *Procedia Engineering* 69:486–491. <https://doi.org/10.1016/j.proeng.2014.03.016>
- Samargandi N (2017) Sector value addition, technology and CO2 emissions in Saudi Arabia. *Renew Sustain Energy Rev* 78(April):868–877. <https://doi.org/10.1016/j.rser.2017.04.056>
- Sandberg M, Klockars K, Wilén K (2019) Green growth or degrowth? Assessing the normative justifications for environmental sustainability and economic growth through critical social theory. *J Clean Prod* 206:133–141. <https://doi.org/10.1016/j.jclepro.2018.09.175>
- Sarikaya M, Güllü A (2015) Multi-response optimization of minimum quantity lubrication parameters using Taguchi-based grey relational analysis in turning of difficult-to-cut alloy Haynes 25. *J Clean Prod* 91:347–357. <https://doi.org/10.1016/j.jclepro.2014.12.020>
- Sato Y (2007) Energy consumption: an environmental problem. *IEEE Trans Electr Electron Eng* 2(1):12–16. <https://doi.org/10.1002/tee.20109>
- Schiederig T, Tietze F, Herstatt C (2012) Green innovation in technology and innovation management - an exploratory literature review. *R and D Management* 42(2):180–192. <https://doi.org/10.1111/j.1467-9310.2011.00672.x>
- Shahzad SJH, Kumar RR, Zakaria M, Hurr M (2017) Carbon emission, energy consumption, trade openness and financial development in Pakistan: a revisit. *Renew Sustain Energy Rev* 70(November 2015):185–192. <https://doi.org/10.1016/j.rser.2016.11.042>
- Shaikh IA, Colarelli O'Connor G (2020) Understanding the motivations of technology managers in radical innovation decisions in the mature R&D firm context: an agency theory perspective. *Journal of Engineering and Technology Management - JET-M*, 55(December 2019). <https://doi.org/10.1016/j.jengtecman.2020.101553>
- Shi K, Wu L (2020) Forecasting air quality considering the socio-economic development in Xingtai. *Sustain Cities Soc* 61(May):102337. <https://doi.org/10.1016/j.scs.2020.102337>

- Shrestha RM, Shrestha RM (2013) Linked references are available on JSTOR for this article: supplemental data for “energy use and greenhouse gas emissions in selected Hindu Kush–Himalayan countries” by. 33(3):33–35
- Smith KR, Frumkin H, Balakrishnan K, Butler CD, Chafe ZA, Fairlie I, Kinney P, Kjellstrom T, Mauzerall DL, McKone TE, McMichael AJ, Schneider M (2013) Energy and human health. *Annu Rev Public Health* 34:159–188. <https://doi.org/10.1146/annurev-publhealth-031912-114404>
- Sohn SY, Jeon J, Han EJ (2015) A new cost of ownership model for the acquisition of technology complying with environmental regulations. *J Clean Prod* 100:269–277. <https://doi.org/10.1016/j.jclepro.2015.03.057>
- Song X, Zhou Y, Jia W (2019) How do economic openness and R&D investment affect green economic growth?—evidence from China. *Resources, Conservation and Recycling* 146(December 2018):405–415. <https://doi.org/10.1016/j.resconrec.2019.03.050>
- Stoknes PE, Rockström J (2018) Redefining green growth within planetary boundaries. *Energy Res Soc Sci* 44(April):41–49. <https://doi.org/10.1016/j.erss.2018.04.030>
- Su CW, Naqvi B, Shao XF, Li JP, Jiao Z (2020) Trade and technological innovation: the catalysts for climate change and way forward for COP21. *J Environ Manage* 269(March):110774. <https://doi.org/10.1016/j.jenvman.2020.110774>
- Sulaiman C, Abdul-Rahim AS (2018) Population growth and CO2 emission in Nigeria: a recursive ARDL approach. *SAGE Open* 8(2). <https://doi.org/10.1177/2158244018765916>
- Tambovceva, T., & Geipele, I. (2012). Environmental management systems experience among Latvian construction companies. *Technological and Economic Development of Economy*, 17(4), 595–610. <https://doi.org/10.3846/20294913.2011.603179>
- Testa F, Rizzi F, Daddi T, Gusmerotti NM, Frey M, Iraldo F (2014) EMAS and ISO 14001: the differences in effectively improving environmental performance. *J Clean Prod* 68:165–173. <https://doi.org/10.1016/j.jclepro.2013.12.061>
- The world bank (2018) Patent applications, Pakistan. <https://data.worldbank.org/indicator/IP.PAT.RESD?locations=PK>
- Tseng ML, Wang R, Chiu ASF, Geng Y, Lin YH (2013) Improving performance of green innovation practices under uncertainty. *J Clean Prod* 40:71–82. <https://doi.org/10.1016/j.jclepro.2011.10.009>
- Tugcu CT, Ozturk I, Aslan A (2012) Renewable and non-renewable energy consumption and economic growth relationship revisited: evidence from G7 countries. *Energy Econ* 34(6):1942–1950. <https://doi.org/10.1016/j.eneco.2012.08.021>
- Umar M, Ji X, Kirikkaleli D, Shahbaz M, Zhou X (2020) Environmental cost of natural resources utilization and economic growth: can China shift some burden through globalization for sustainable development? *Sustain Dev* 28(6):1678–1688. <https://doi.org/10.1002/sd.2116>
- Vazquez-Brust D, Smith AM, Sarkis J (2014) Managing the transition to critical green growth: the “green growth state.” *Futures* 64:38–50. <https://doi.org/10.1016/j.futures.2014.10.005>
- Vo DH, Vo AT (2021) Renewable energy and population growth for sustainable development in the Southeast Asian countries. *Energy Sustain Soc* 11(1):1–15. <https://doi.org/10.1186/s13705-021-00304-6>
- Wagner M (2010) The role of corporate sustainability performance for economic performance: a firm-level analysis of moderation effects. *Ecol Econ* 69(7):1553–1560. <https://doi.org/10.1016/j.ecolecon.2010.02.017>
- Wakeel M, Chen B, Jahangir S (2016) Overview of energy portfolio in Pakistan. *Energy Procedia* 88:71–75. <https://doi.org/10.1016/j.egypro.2016.06.024>
- Walker RM, Damanpour F, Devece CA (2011) Management innovation and organizational performance: the mediating effect of performance management. *J Public Adm Res Theory* 21(2):367–386. <https://doi.org/10.1093/jopart/muq043>
- Wang X, Shao Q (2019) Non-linear effects of heterogeneous environmental regulations on green growth in G20 countries: evidence from panel threshold regression. *Sci Total Environ* 660:1346–1354. <https://doi.org/10.1016/j.scitotenv.2019.01.094>
- Wang Z, Zhu Y (2020) Do energy technology innovations contribute to CO2 emissions abatement? A spatial perspective. *Sci Total Environ* 726:138574. <https://doi.org/10.1016/j.scitotenv.2020.138574>
- Wang P, Liu Q, Qi Y (2014) Factors influencing sustainable consumption behaviors: a survey of the rural residents in China. *J Clean Prod* 63:152–165
- Wang Y, Liu X, Sun X, Wang B (2020) Energy saving, GHG abatement and industrial growth in OECD countries: a green productivity approach. *Energy* 194:116833. <https://doi.org/10.1016/j.energy.2019.116833>
- Wang KH, Umar M, Akram R, Caglar E (2021) Is technological innovation making world “greener”? An evidence from changing growth story of China. *Technological Forecasting and Social Change* 165(July 2020):120516. <https://doi.org/10.1016/j.techfore.2020.120516>
- World health Organization (2020) Environmental health. World Health Organization, Regional Office for Eastern Mediterranean. <http://www.emro.who.int/pak/programmes/environmental-health.html>
- Wu W (2017) Institutional quality and air pollution: international evidence. *Int J Bus Econ* 16(1):49–74
- Xu J, Li Y (2018) Grey incidence analysis model of classification variables and its application on innovation & entrepreneurship education in Jiangsu. *J Grey Syst* 30(1):123–128
- Xu Y, Du J, Shahzad F, Li X (2021) Untying the influence of green brand authenticity on electronic word-of-mouth intention: a moderation–mediation model. *Frontiers in Psychology*, 12(September). <https://doi.org/10.3389/fpsyg.2021.724452>
- Zhai X, An Y (2021) The relationship between technological innovation and green transformation efficiency in China: an empirical analysis using spatial panel data. *Technology in Society* 64(September 2020) 101498. <https://doi.org/10.1016/j.techsoc.2020.101498>
- Zhang M, Ding S, Pang J, Wang W (2021) The effect of indirect household energy consumption on PM 2.5 emission in China: an analysis based on CLA method. *J Environ Manage* 279(April 2020):111531. <https://doi.org/10.1016/j.jenvman.2020.111531>
- Zhang S, Hu T, Li J, Cheng C, Song M, Xu B, Baležentis T (2019) The effects of energy price, technology, and disaster shocks on China’s energy–environment–economy system. *J Clean Prod* 207:204–213. <https://doi.org/10.1016/j.jclepro.2018.09.256>
- Zhao R, Liu Y, Zhang N, Huang T (2017) An optimization model for green supply chain management by using a big data analytic approach. *J Clean Prod* 142:1085–1097
- Zhicai Z, Li C (2018) Analysis on decision-making model of plan evaluation based on grey relation projection and combination weight algorithm. *J Syst Eng Electron* 29(4):789–796
- Ziegler A, Seijas Nogareda J (2009) Environmental management systems and technological environmental innovations: exploring the causal relationship. *Res Policy* 38(5):885–893. <https://doi.org/10.1016/j.respol.2009.01.020>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.